Hanrong Zhang

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Education

University of Illinois Chicago — **Aug. 2025 – Aug. 2028 (Expected)**
PhD Student in Computer Science Advisor: Prof. Philip S. Yu — *Chicago, USA*

- **Tencent Project Up Scholarship** (4 recipients worldwide)

Zhejiang University — **Sep. 2022 – Mar. 2025**
MEng Computer Engineering Ranking: 1/82 — *Hangzhou, China*

- **National Scholarship for Graduate Students** ×**2** (Top 0.2%)
- Outstanding Graduate in Zhejiang Province & Zhejiang University

University of Leeds — **Sep. 2018 – Jun. 2022**
BSc Computer Science First Class Honours — *Leeds, United Kingdom*

- Best Student Overall (1/300); Best Student in Computer Science (1/75)
- First-class Full-ride Scholarship (1/75)

Southwest Jiaotong University — **Sep. 2018 – Jun. 2022**
BEng Computer Science and Technology Ranking: 1/75 GPA: 92/100 — *Chengdu, China*

- National Scholarship for Undergraduate Students (Top 0.2%)
- Outstanding Graduate in Sichuan Province

Internship

Google — **May 2026 – Present**
Student Researcher, Data Synthesis for Gemini and Long-Horizon GUI Agents — *Mountain View, California, US*

- Synthesize large-scale, diverse **task queries and GUI trajectories** to strengthen **Gemini**'s GUI-agent capabilities.
- Build **Mobile Skills Bench (MSB)**, a benchmark for long-horizon interactive mobile agents. *(Under review at ICLR 2027)*
  - Synthesize verifiable, multi-stage cross-app tasks, paired with an interactive user simulator.
  - Benchmark frontier models on long-horizon tasks under both pure-GUI and hybrid (GUI + MCP + CLI) execution.
  - Design an efficient **skill organization** for mobile agents and quantify the value of textual vs. multimodal skills.
- Build **Android-based mock apps** with resettable backend state and verifiable outcomes for internal Android-agent evaluation.

Alibaba Group — **May 2025 – Aug. 2025**
Research Intern, RL for Human-Agent Multi-turn Interaction — *Hangzhou, Zhejiang, China*

- Build scalable simulation environments for multi-turn human–agent tool-call interactions (e.g., retail and airline services), tackling sparse rewards, scarce data, and environment instability.
- Propose a **Tool Dependency Graph** that models tool constraints to synthesize high-quality multi-turn tool-call dialogues with ground-truth trajectories for verifiable rewards; train agents on these trajectories via multi-turn **GRPO**, improving long-horizon tool use in human–agent interaction.

Selected Publications and Preprints

*Full publication list is available on Google Scholar. * denotes Equal Contribution. † denotes Corresponding Author.*

[1] **H. Zhang**, J. Huang, K. Mei, Y. Yao, Z. Wang, C. Zhan, H. Wang, Y. Zhang, *Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents*, **ICLR 2025**. [Code] [Website] LLM Agent Security ~650 citations · 300+ GitHub stars

[2] **H. Zhang**, S. Fan, H. P. Zou, Y. Chen†, Z. Wang, J. Zhou, C. Li, W.-C. Huang, Y. Yao, K. Zheng, X. Liu, X. Li, P. S. Yu, *CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification*, **COLM 2026**. [Code] Agent Self-Evolution ~130 citations

[3] **H. Zhang**, Z. Wang, T. Han, M. Jin, C. Zhan, M. Du, H. Wang†, S. Ma, *Invisible Backdoor Attack in Self-supervised Learning*, **CVPR 2025**. [Code] AI Security

[4] Z. Wu*, **H. Zhang***†, F. Lin, W. Xu, X. Xu, Y. Chen, H. P. Zou, S. Chen, W. Zhang, X. Liu, P. S. Yu, H. Wang†, *GAM: Hierarchical Graph-based Agentic Memory for LLM Agents*, **ACL 2026**; ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems Oral. Agent Memory

[5] **H. Zhang**, Y. Chen†, S. Fan, . . . , D. Song, P. S. Yu, X. Liu, *Scaling LLM Agent Learning with Data Synthesis: A Comprehensive Survey*, **Preprint**. Agent Learning & Data Synthesis

[6] Z. Yan, D. Song, **H. Zhang**, W. Liang, Y. Zhang, Y. Dai, L. He, P. S. Yu, R. Xu, X. Li, L. Sun, *OpenSkill: Open-World Self-Evolution for LLM Agents*, **EMNLP 2026**. [Code] Agent Self-Evolution

[7] **H. Zhang**, Y. Yao, Z. Wang, J. Su, M. Li, P. Peng, H. Wang, *Class Incremental Fault Diagnosis under Limited Fault Data via Supervised Contrastive Knowledge Distillation*, **IEEE Transactions on Industrial Informatics**. Continual Learning IF 12.3 · SCI Q1 · Top Journal

[8] Y. Yao*, **H. Zhang***, F. Lin*, Z. Jin, X. Li, H. Wang, Y. Chi, *From Uncertainty to Clarity: Uncertainty-Guided Class-Incremental Learning for Limited Biomedical Samples via Semantic Expansion*, **IEEE Journal of Biomedical and Health Informatics**. Continual Learning IF 7.7 · SCI Q1

[9] X. Wang*, **H. Zhang***, X. Qiao, K. Ma, S. Tao, P. Peng, H. Wang, *Generalized Out-of-distribution Fault Diagnosis (GOOFD) via Internal Contrastive Learning*, **IEEE Transactions on Industrial Informatics**. OOD Detection IF 12.3 · SCI Q1 · Top Journal

Other co-authored papers: NeurIPS 2026 ×2, ICML 2026, ACL 2026 ×2, EMNLP 2026 ×3, TMLR ×2, etc.

Research Experience

Agent Security Bench (ASB) **ICLR 2025, First Author**

- **Motivation:** LLM agents extend LLM reasoning into real-world action: by calling tools, updating memory, and executing multi-step plans, they directly change external environments, making agent security more critical than standard AIGC safety; yet prior evaluations lacked a unified benchmark spanning prompts, tools, memory, and planning.
- **Contribution:** Introduce **the first fundamental agent security evaluation benchmark**, with 10 scenarios, 10 agents, 400+ tools, 27 attack/defense methods, 7 metrics, and Net Resilient Performance (NRP) for balancing utility and security.
- **Results:** Across 13 LLM backbones and 90,000+ test cases, reveal that current LLM agents remain highly vulnerable, with the highest average attack success rate reaching 84.30%, while existing defenses offer limited protection.

Towards Self-Evolving Skills in LLM Agents **CoEvoSkills (COLM 2026, First Author) & OpenSkill (EMNLP 2026)**

- **Motivation:** Agent skills can boost performance on professional tasks, but we identify a **human–machine misalignment**: human-curated skills are costly, and skills humans find helpful are not necessarily useful to agents, sometimes even hurting performance in SkillsBench domains; agents should therefore author their own skills.
- **Contribution:** Propose **CoEvoSkills**, which co-evolves multi-file skill packages with a surrogate verifier in closed-world settings, and **OpenSkill**, which self-evolves skills via open-world knowledge retrieval.
- **Results: CoEvoSkills** achieves a 71.1% average pass rate on SkillsBench (+40.5pp over no skills and +17.6pp over human-curated skills); **OpenSkill** reaches 43.6% with Claude Opus 4.6 and 42.1% with GPT 5.2, within 1–3pp of human-curated skills and about 9pp above the best closed-world baseline.

Hierarchical Graph-based Agentic Memory (GAM) **ACL 2026, Co-First & Corresponding Author**

- **Motivation:** Long-term agent memory must reconcile fast contextual adaptation with stable knowledge retention: stream-based memories update quickly but lack write isolation, causing memory loss and semantic drift, whereas static graph memories preserve structure but adapt poorly to fluid open-domain dialogue.
- **Contribution:** Propose **GAM**, a hierarchical graph memory framework that decouples memory encoding from consolidation: an Event Progression Graph isolates ongoing dialogue events, and semantic-boundary triggers consolidate completed episodes into a Topic Associative Network for stable long-term knowledge.
- **Results: GAM** outperforms six memory baselines on both LoCoMo and LongDialQA.

Open-Source Projects

Dr. Claw [Code] 1.1k+ GitHub stars · 130+ forks **EMNLP 2026 · Second Author · Core Contributor**

- Build an **Auto Research** workspace that orchestrates agentic pipelines from research ideation and literature/code survey to experiment execution, analysis, paper writing, and presentation generation.
- Integrate multi-source research news tracking, 100+ research skills, project/session management, and multi-agent backends into a full-stack research IDE.

Invited Talks

Scaling GUI Agents with Synthetic Data and Environments: From Training to Evaluation **NVIDIA · Oct. 2026**

Towards Self-Evolving AI Agent Skills **Alibaba Group · Jun. 2026**

How to Evaluate AI Agent Security? **National University of Singapore · Apr. 2025**

Academic Service

Reviewer ICLR, ICML, NeurIPS, CVPR, ECCV, KDD, AAAI, MICCAI; TPAMI, TNNLS, TMLR, etc.